

SRI KRISHNA KALYAN BANDARU

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ABOUT ME

ML Engineer with a Master of Artificial Intelligence and Machine Learning from the University of Adelaide and a background spanning mechanical engineering, mechatronics, and embedded systems. End-to-end AI solutions delivered — from dataset construction and model development through to production deployment and stakeholder communication. Specialising in computer vision, edge AI, and autonomous systems, with hands-on experience deploying models to NVIDIA Jetson hardware, building FastAPI inference services, and navigating full MLOps lifecycles. Cross-disciplinary by design: hardware intuition meets production ML engineering.

EDUCATION

Master of Artificial Intelligence & Machine Learning	Dec 2025
University of Adelaide GPA: 6.1 Focus: Computer Vision, Deep Learning, NLP, MLOps	
Master of Mechatronics Engineering	2022
University of Rome 'Tor Vergata' Grade: 106/110 Focus: Control Systems, Embedded AI, MATLAB/Simulink	
Bachelor of Technology in Mechanical Engineering	2018
KL University, India CGPA: 8.89/10, First Class Distinction Focus: Engineering Fundamentals, C Programming	

TECHNICAL SKILLS

Languages	Python, SQL, C/C++, Java, MATLAB/Simulink, Git
Computer Vision	YOLO v5-v12, Detectron2, Mask R-CNN, OpenCV, Instance Segmentation, Roboflow
ML Frameworks	PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face, Ultralytics, BERT, Transfer Learning
Deployment & Cloud	FastAPI, Docker, Streamlit, REST APIs, AWS, Azure, CI/CD, Linux
Robotics & Edge AI	ROS2 (Humble), Gazebo, NVIDIA Jetson Nano/Orin, STM32, ArduPilot SITL, MAVROS
Data & Analytics	Pandas, NumPy, Matplotlib, Tableau, BigQuery, Statistical Analysis, EDA

EXPERIENCE

Research Assistant — Auto-ID Lab, University of Adelaide Feb 2025 – Aug 2025
Adelaide, Australia — Autonomous Systems & Computer Vision

- Led full migration of UAV codebase from ROS1 to ROS2 — restoring compatibility with modern hardware and enabling deployment on current drone platforms
- Built complete Gazebo indoor warehouse simulation environment for safe AI algorithm validation — reducing hardware risk and accelerating research iteration cycles
- Containerised and deployed full inference stack onto NVIDIA Jetson Nano/Orin — achieving stable, production-ready autonomous drone operations on edge hardware
- Evaluated YOLO architectures (YOLOv5s through YOLOv12s) on VisDrone aerial datasets, achieving **mAP50 of 0.642** for real-time person and vehicle detection; integrated SuperPoint feature tracking, increasing visual features from 5 to **~2,000 per frame**

ML Engineer Intern — Proto Corp Pvt Ltd Aug 2024 – Feb 2025
Remote, Australia — Computer Vision & MLOps

- Collected, cleaned, and annotated **3,500+ construction defect images** at pixel-level precision across 5 categories — enabling the first production-ready CV pipeline for infrastructure inspection
- Evaluated YOLOv8m-seg, YOLOv9c-seg, YOLOv11m-seg, and Mask R-CNN through rigorous comparative testing; selected YOLOv8m-seg for production based on accuracy and inference speed
- Built FastAPI inference backend and Streamlit dashboard enabling non-technical stakeholders to run real-time defect segmentation — removing bottlenecks in the quality inspection workflow
- Maintained continuous dataset versioning, annotation pipelines, and retraining workflows across the full ML lifecycle using Roboflow and Confluence

Project Engineer — Wipro Technologies May 2018 – Sep 2019
Hyderabad, India — IT Operations & SAP PM Support

- Managed plant maintenance operations for UK industrial clients using SAP PM; led a team of four to consistently meet SLA targets and minimise equipment downtime
- Identified recurring escalation patterns through root-cause documentation, reducing repeated incidents and improving team knowledge base quality

KEY PROJECTS

Autonomous UAV Object Detection System — ROS2, YOLOv12s, NVIDIA Jetson, EGO-Planner, Gazebo, VINS-Fusion
Developed modular ROS2 architecture combining EGO-Planner obstacle avoidance with YOLO real-time object detection for disaster response applications. Validated via ArduPilot SITL hardware-in-the-loop simulation; achieved **mAP50 of 0.642** on VisDrone and **~2,000 features/frame** with SuperPoint feature tracking.

Embedded Neural Network on STM32 (On-Device AI) — TensorFlow Lite, STM32CubeMX, X-CUBE-AI, C/C++, Keras
Applied model compression and weight quantisation (~3× reduction) to deploy a CNN directly onto STM32H747I-DISCO micro-controller, eliminating cloud dependency. Achieved **97% accuracy** across 18 food categories at **11.2 FPS** fully on-device with 81ms inference latency.

Alzheimer's Disease Diagnosis from MEG Signals — PyTorch, AlexNet, ResNet, GoogLeNet, SVM, Scikit-learn
Engineered MEG neurological time-series into 2D representations and applied transfer learning across AlexNet, ResNet, and GoogLeNet with leave-one-patient-out cross-validation. Achieved **81% accuracy** and **AUC of 0.74** for early-stage Alzheimer's detection.